PRIYANKA PILLAI

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PROFESSIONAL EXPERIENCE

ROCKET MORTGAGE **May 2026 – Aug 2026**
Software Engineering Intern • Detroit, MI
- Shipped production features for a customer-facing financial platform using Angular, TypeScript, C#, and .NET, contributing to faster loan-processing workflows within an agile sprint cycle.
- Accelerated feature delivery by integrating AI-assisted development tools including GitHub Copilot, cutting debugging time and improving code quality on mortgage transaction systems.

COLLINS AEROSPACE **Sep 2022 – Dec 2024**
Lead Software Developer
- **System Design & APIs:** Architected and delivered RESTful APIs integrating federated aviation systems with external platforms, improving interoperability and reducing manual processes by 40%.
- **Scalable Infrastructure:** Deployed cloud-native microservices on AWS and Azure supporting 50K+ active users; engineered low-latency endpoints with real-time monitoring and fault tolerance.
- **CI/CD & Reliability:** Established CI/CD pipelines with Jenkins and GitLab, reducing deployment errors by 25% and cutting release cycles from two weeks to under five days.
- Eliminated performance bottlenecks and concurrency issues in multithreaded systems via advanced profiling techniques, improving uptime by 15%.

TATA CONSULTANCY SERVICES **Oct 2020 – Sep 2022**
System Engineer
- Developed scalable Spring Boot microservices processing 1M+ telemetry signals daily with high fault tolerance; applied algorithmic optimizations that eliminated throughput bottlenecks and reduced processing delays.
- Engineered RESTful APIs on a microservices architecture, cutting downtime event reporting latency from 15 minutes to under 2 minutes and improving system availability by 93%.
- Diagnosed and resolved memory leaks, race conditions, and concurrency bugs; delivered across 6 agile Scrum sprints with less than 2% defect rate.

PROJECTS

Text2SQL – NL-to-SQL Pipeline with RAG & LLM Validation (Python, TypeScript, FastAPI, Next.js, SQL, AWS)
- Engineered an end-to-end Text-to-SQL system with a FAISS-based RAG retrieval pipeline over 8,659 question–SQL pairs; implemented a Steiner tree algorithm for schema-grounded join-path construction across complex multi-table schemas.
- Achieved 74.9% execution accuracy on Spider and 62.1% on CoSQL with the full Qwen2.5-Coder + RAG + Repair + Semantic Validation pipeline; failure mode analysis pinpointed filter conditions as the dominant error class (50%).

Intelligent Health Monitoring & Heart Attack Prediction System (Python, AWS S3, EMR, SageMaker, Lambda, SNS, Athena, Spark)
- Architected an end-to-end distributed ML pipeline on AWS: streamed simulated wearable vitals into S3, processed 7-day patient data via Apache Spark on EMR, and trained an XGBoost classifier on SageMaker to predict heart attack risk.
- Launched a real-time SageMaker inference endpoint; automated high-risk patient alerts via AWS Lambda and SNS; surfaced query results through Athena SQL, covering the full data engineering, model training, and alerting lifecycle.

EDUCATION

ARIZONA STATE UNIVERSITY **Jan 2025 – Dec 2026**
Master of Science, Information Technology

BASAVESHWAR ENGINEERING COLLEGE **Aug 2016 – Aug 2020**
Bachelor of Engineering, Computer Science and Engineering

TECHNICAL SKILLS

- **Languages:** Python, Java, C#, TypeScript, JavaScript, SQL, C, C++, HTML5, CSS3
- **Frameworks & Libraries:** Angular, .NET, React, Node.js, Spring Boot, FastAPI, Next.js, Django
- **Systems & Cloud:** Distributed Systems, Microservices, AWS, Azure, Docker, Kubernetes, CI/CD, REST APIs
- **Databases & Tools:** PostgreSQL, MySQL, MongoDB, SQLite, FAISS, Git, Jenkins, Postman
- **CS Fundamentals:** Data Structures & Algorithms, System Design, OOP, Concurrency, Agile/Scrum